# On the Nearest Special Unitary Matrix

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## Abstract

This paper derives the optimal solution for finding the nearest special unitary matrix to a general square complex matrix under the Frobenius norm. We present both an exact analytical characterization and a faster numerical method. The results are validated experimentally and compared with common ersatz approaches.


## 1 Introduction and Background

Matrix nearness problems arise in many mathematical and engineering applications. The basic task is to find the closest constrained matrix $Q$ to a given matrix $M$ under a chosen distance measure. For example, in rotation estimation, one may start with a noisy matrix and project it onto the set of valid rotations. In quantum computing, a learned or approximate operator may need to be replaced by the closest physically valid evolution. In optimization algorithms, numerical drift from composite updates may cause a candidate to leave the feasible set, requiring a projection step before continuing.

A convenient and widely used choice of matrix distance is the squared Frobenius norm (Higham, 1989):

$$||M||_F^2 = Tr(M^H M) \tag{1}$$

where $Tr(\cdot)$ denotes the trace and $^H$ denotes the conjugate transpose. This choice is attractive because it is easy to compute, unitarily invariant, and differentiable. Furthermore, the squared norm is chosen over the norm because it is equivalent as an optimizer while leading to cleaner algebra. With this metric, we can phrase our matrix nearness problems as:

$$\min_{Q \in S} ||M - Q||_F^2 = Tr((M - Q)^H (M - Q)) \tag{2}$$

where $Q$ is constrained to lie in a feasible set $S$.

Many choices of $S$ have been well-studied over the years and admit simple analytical solutions to Eq. (2) (Higham, 1989). Orthogonal matrices for instance are real square matrices $Q \in O(n) \subset \mathbb{R}^{n\times n}$ satisfying $Q^T Q = I$, where $^T$ denotes transpose. For a real matrix $M \in \mathbb{R}^{n\times n}$, the minimizer of Eq. (2) over $O(n)$ is the orthogonal factor of the polar decomposition of $M$. Equivalently, if $M$ is written via the polar decomposition or the singular value decomposition (SVD), then (Keller, 1975):

$$M = AP = U\Sigma V^T \tag{3}$$

$$Q = A = UV^T \tag{4}$$

Here, $A$, $U$, and $V^T$ are orthogonal matrices, $P$ is symmetric positive semidefinite, and $\Sigma$ is diagonal with nonnegative entries. Thus the orthogonal nearness problem has an exact projection formula obtained directly from standard matrix factorizations.

Special orthogonal matrices are a subset of orthogonal matrices that impose the additional constraint of having determinant one. That is, $Q \in SO(n)$ if $Q \in O(n)$ and $\det(Q) = 1$. In this case, the nearest special orthogonal matrix remains explicit and is a simple correction of the orthogonal solution (Wahba, 1965):

$$Q = UWV^T \tag{5}$$

$$W = diag(1,\ 1,\ \ldots,\ \det(U) * \det(V^T))$$

where $diag(\cdot)$ denotes a diagonal matrix formed from the listed entries. The correction flips only the final singular direction when needed, so the rest of the optimal orthogonal projection is preserved. Note that Eq. (5) assumes the SVD is defined such that that singular values are in nonincreasing order.

Unitary matrices generalize orthogonal matrices to the complex setting. They are square matrices $Q \in U(n) \subset \mathbb{C}^{n\times n}$ satisfying $Q^H Q = I$. For a complex matrix $M \in \mathbb{C}^{n\times n}$, the minimizer of Eq. (2) over $U(n)$ is again the polar factor. Equivalently, if $M$ is written via the polar decomposition or the SVD, then (Keller, 1975):

$$M = AP = U\Sigma V^H \tag{6}$$

$$Q = A = UV^H \tag{7}$$

In this case, $A$, $U$, and $V^H$ are unitary, $P$ is Hermitian positive semidefinite, and $\Sigma$ is diagonal with nonnegative entries. Just as in the orthogonal case, the nearest feasible matrix is obtained by keeping the unitary part of the factorization.

Special unitary matrices are the complex analogue of $SO(n)$. Due to their deep connection to physics and rotations, they are increasingly applied to engineering areas such as robotics (Chandrasekhar, 2026), deep learning (Romiti, 2026) and quantum computing (Wiersema et al., 2024). Formally, a matrix belongs to $SU(n)$ if $Q \in U(n)$ and $\det(Q) = 1$. At first glance, one might expect the nearest special unitary matrix problem to mirror the special orthogonal case and amount to a simple correction of the unitary solution. However, as we later show, the determinant constraint is more nuanced in the complex setting because it fixes a continuous phase rather than a discrete sign, making the optimization problem more involved.

### 1.1 Contributions

$$\min_{Q\in SU(n)} ||M - Q||_F^2 = Tr((M - Q)^H (M - Q)) \tag{8}$$

We derive the complete optimal solution to the problem of finding the nearest special unitary matrix to a general complex matrix in the Frobenius sense (Eq. (8)). Our analysis shows that unlike the previously mentioned classical matrix nearness problems, the special unitary case is fundamentally more complex and requires a more careful optimization treatment. In addition to an analytical solution, we develop a faster numerical method, discuss important special cases, and validate our methods through numerical experiments.

## 2 Naive Solutions

Before deriving the exact solution, it is useful to examine two natural heuristics for projecting onto $SU(n)$. Both are easy to motivate from previously known matrix nearness results, and both produce valid special unitary matrices. However, neither is necessarily optimal for Eq. (8) on a general complex matrix.

### 2.1 Naive Solution I: Correction to Unitary Solution

The first heuristic starts from the nearest unitary matrix in Eq. (7) and then applies a global phase correction so that the determinant becomes one:

$$Q = \Big(\det(A)^{-\frac{1}{n}}\Big)A = \Big(\det(UV^H)^{-\frac{1}{n}}\Big)UV^H \tag{9}$$

It is immediately seen that $Q \in SU(n)$. From another perspective, this construction is equivalent to first rescaling $M$ so that its determinant is real and nonnegative, and then projecting onto $U(n)$ which will coincide in this case with $SU(n)$ (Chandrasekhar, 2026). If $M$ is viewed projectively (i.e. as an element of $PGL(n, \mathbb{C})$), then this interpretation is reasonable. The method returns the closest special unitary matrix over all nonzero scalings of $M$ (see Appendix A.3.1). However, our objective in Eq. (8) is posed for a fixed matrix $M \in GL(n, \mathbb{C})$. Under that objective, scaling changes the problem, so this heuristic does not in general return the nearest special unitary matrix to $M$ itself.

### 2.2 Naive Solution II: Analogue of Special Orthogonal Solution

The second heuristic is a direct complex analogue of the special orthogonal solution in Eq. (5). It modifies only the singular direction associated with the smallest singular value:

$$Q = UW'V^H \tag{10}$$

$$W' = diag(1,\ 1,\ \ldots,\ (\det(U) * \det(V^H))^*) \tag{11}$$

where $^*$ denotes complex conjugation. Like the special orthogonal formula, this construction enforces the determinant constraint by correcting a single factor while leaving the remaining singular directions unchanged. Although intuitive as a method, our experiments will show that this approach generally yields suboptimal results.

## 3 Optimal Solution

We now derive the structure of the optimal solution to the nearest special unitary matrix problem. Starting from Eq. (8), we rewrite the objective as follows:

$$\min_{Q \in SU(n)} ||M - Q||_F^2 = \min_{Q \in SU(n)} Tr(M^H M) - 2\Re(Tr(Q^H M)) + Tr(Q^H Q) \tag{12}$$

where $\Re(\cdot)$ denotes the real part (see Appendix A.1 for details). Since $Q^H Q = I$ for any $Q \in SU(n)$, the first and third terms in Eq. (12) are constant with respect to $Q$. Therefore, the problem is equivalent to:

$$\max_{Q \in SU(n)} 2\Re(Tr(Q^H M)) \tag{13}$$

Using the method of Lagrange multipliers to optimize over the unitary and determinant constraints, the first-order optimality conditions yield the stationary relation (see Appendix A.2 for derivation):

$$Q^H M = \Lambda + \lambda^* I \tag{14}$$

where $\Lambda$ is a Hermitian matrix and $\lambda$ is a complex scalar. This condition has two immediate consequences. First, since $\Lambda$ is Hermitian, its eigenvalues are real, so adding $\lambda^* I$ shifts them all by the same complex scalar. Thus, the eigenvalues of $Q^H M$ all share the same imaginary component. Second, $\Lambda + \lambda^* I$ is normal, and therefore, $Q^H M$ admits a unitary diagonalization of the form:

$$Q^H M = R \Omega R^H \tag{15}$$

where $R$ is unitary and $\Omega = diag(\mu_1, \ldots, \mu_n)$ is diagonal. Next, we express the singular value decomposition of $M$ as $M = U\Sigma V^H$ with singular values $\sigma_1 \geq \cdots \geq \sigma_n \geq 0$. Combining Eq. (15) with the SVD of $M$ gives:

$$(Q^H M)^H (Q^H M) = M^H M = V \Sigma^2 V^H = R \Omega^H \Omega R^H \tag{16}$$

Therefore, $\Omega^H \Omega$ and $\Sigma^2$ are equivalent up to permutation, and we may choose an ordering so that:

$$|\mu_i| = \sigma_i, \qquad i = 1, \ldots, n. \tag{17}$$

where $|\cdot|$ denotes complex magnitude. Writing $\mu_i = \sigma_i z_i$ with $z_i \in \mathbb{C}$ and $|z_i| = 1$, the objective in Eq. (13) becomes:

$$2\Re(Tr(Q^H M)) = 2\Re(Tr(R\Omega R^H)) = 2\Re(Tr(\Omega)) = 2\sum_{i=1}^{n} \sigma_i \Re(z_i). \tag{18}$$

We now rewrite the determinant constraint in terms of the phases $z_i$:

$$\det(Q^H M) = \det(Q^H)\det(M) = \det(M) = \det(U\Sigma V^H) = \det(U)\det(\Sigma)\det(V^H) \tag{19}$$

$$= \det(R)\det(\Omega)\det(R^H) = \det(\Omega) = \det(\Sigma)\prod_{i=1}^{n} z_i \tag{20}$$

Assuming $M$ is full-rank, it follows that:

$$\prod_{i=1}^{n} z_i = \det(U)\det(V^H) \tag{21}$$

Hence, the original problem reduces to the following optimization over $z_i$:

$$\max_{z_i} \; 2\sum_{i=1}^{n} \sigma_i \Re(z_i) \quad \text{s.t.} \quad |z_i| = 1, \; \prod_{i=1}^{n} z_i = \det(U)\det(V^H) \tag{22}$$

This optimization can be viewed intuitively as tweaking a set of "knobs" represented by $z_i$ in order to maximize the sum of their real parts (weighted by $\sigma_i$), while constraining the cumulative sum of knob angles to a specific angle. The objective can be written equivalently in matrix form as:

$$\max_{Z \in U(n)} 2\Re(Tr(Z\Sigma)) \quad \text{s.t.} \quad Z = diag(z_1, \ldots, z_n), \;\; \det(Z) = \det(U)\det(V^H), \tag{23}$$

Given any feasible optimizer $Z$ of Eq. (23), we relate it back to the original problem through:

$$Tr(Z\Sigma) = Tr((V^H V)Z(U^H U)\Sigma) = Tr(VZU^H(U\Sigma V^H)) = Tr((VZU^H)M) \tag{24}$$

Since $VZU^H$ is special unitary, it is a valid maximizer of Eq. (13). Identifying it with $Q^H$ yields the solution:

$$Q = UZ^* V^H \tag{25}$$

The next section details how to find the optimal choice of phases $\{z_1, \ldots, z_n\}$.

### 3.1 Analytical Approach

There are several ways to derive an analytical solution. We present one relatively efficient approach below. To solve Eq. (22), let $\rho := \det(U)\det(V^H)$ and write each phase variable as:

$$z_i = a_i + b_i i, \qquad a_i^2 + b_i^2 = 1 \tag{26}$$

where $a_i, b_i \in \mathbb{R}$. With this parametrization, the reduced problem becomes:

$$\max_{a_i, b_i} \; 2\sum_{i=1}^{n} \sigma_i a_i \quad \text{s.t.} \quad a_i^2 + b_i^2 = 1, \; \prod_{i=1}^{n}(a_i + ib_i) = \rho \tag{27}$$

The first-order stationarity conditions imply that the imaginary parts are coupled through the singular values. More precisely, any stationary point satisfies:

$$\sigma_i b_i = \sigma_j b_j, \qquad i \neq j \tag{28}$$

Thus, assuming $M$ is nonsingular, we can write:

$$b_i = \frac{\sigma_n}{\sigma_i} b_n, \qquad i < n \tag{29}$$

We rewrite the determinant constraint in Eq. (21) to isolate the final phase variable:

$$z_n = \rho \prod_{i=1}^{n-1} z_i^* \tag{30}$$

where the conjugate appears because each $z_i$ has unit modulus. Taking imaginary parts gives a single scalar constraint:

$$b_n = \Im\left(\rho \prod_{i=1}^{n-1}(a_i - b_i i)\right) \tag{31}$$

where $\Im(\cdot)$ denotes the imaginary part. Substituting Eq. (29) into the unit-norm constraints gives:

$$a_i^2 = 1 - \frac{\sigma_n^2}{\sigma_i^2} b_n^2, \qquad i < n \tag{32}$$

Similarly, substituting Eq. (29) into Eq. (31) gives a polynomial equation in $a_1, \ldots, a_{n-1}$ and $b_n$ that is linear in each $a_i$ individually.

For each $i < n$, we now apply an iterative elimination procedure that reduces the current equation to the form:

$$c_{i,1} a_i = -c_{i,0} \tag{33}$$

where $c_{i,1}$ and $c_{i,0}$ are polynomials in the remaining variables. This reduction uses Eq. (32) to eliminate higher powers of $a_i$, leaving a linear $a_i$ term for odd powers and a constant term (with respect to $a_i$) for even powers. Squaring both sides of Eq. (33) and substituting for $a_i^2$ gives:

$$c_{i,1}^2 \left(1 - \frac{\sigma_n^2}{\sigma_i^2} b_n^2\right) - c_{i,0}^2 = 0 \tag{34}$$

which removes $a_i$ entirely. Repeating this elimination for $i = 1, \ldots, n-1$ leaves a single univariate polynomial equation in $b_n$:

$$f(b_n) = 0 \tag{35}$$

The resulting polynomial $f$ provides an analytical characterization of all stationary points of the objective in Eq. (22). Its degree grows rapidly with $n$, namely $2^{n-1}(n-1)$ (see Appendix A.3.2). However, one advantage of this particular construction is that when $n$ is even, the polynomial can be written as $f(b_n) = g(b_n^2)$, reducing the effective degree to $2^{n-2+(n \bmod 2)}(n-1)$ in general (see Appendix A.3.3).

Once a root $b_n$ is obtained, the remaining quantities are recovered algebraically. First, the real parts $a_i$ for $i < n$ are recovered by backsubstitution into $c_{i,0}, c_{i,1}$ and solving Eq. (33). If $c_{i,1} = 0$, the positive branch of Eq. (32) can be used instead (see Appendix A.3.5). The imaginary parts then follow directly from Eq. (29). Finally, the last variable is recovered from the real part of Eq. (30):

$$a_n = \Re\left(\rho \prod_{i=1}^{n-1} (a_i - b_i i)\right) \tag{36}$$

This yields a candidate solution $\{z_1, \ldots, z_n\}$. Among the resulting candidates, the best optimizer is the one that maximizes Eq. (22). Note that the sign of the maximizing $b_n$ agrees with $\operatorname{sign}(\Im(\rho))$ for $\Im(\rho) \neq 0$ (see Appendix A.3.4) which can help filter bad candidates. The corresponding optimal matrix is finally obtained from Eq. (25).

### 3.2 Numerical Approach

The analytical approach gives an exact method, but the degree of the univariate polynomial grows quickly with dimension. In addition, the computation becomes numerically unstable rather quickly, even for dimensions as low as $n = 4$. In fact, many higher degree coefficients in higher dimensions can become exceedingly small (e.g. $< 10^{-50}$). Instead of explicitly forming this polynomial, we can exploit the problem structure to numerically approximate the solution.

The stationarity relation in Eq. (29) shows that once one imaginary component is known, the remaining imaginary components are determined as well, and their real components follow subsequently from the unit constraints. As a result, we can easily parameterize the search with a single variable. We choose $b_1$ because the singular values are ordered as $\sigma_1 \geq \cdots \geq \sigma_n$, and therefore:

$$a_1 \geq a_i, \qquad i > 1 \tag{37}$$

by the rearrangement inequality, so $|b_1|$ is expected to be relatively small and limited in its domain. Empirically, we observe that for large $n$ or high condition numbers of $M$, the optimal solution often has $a_1 \approx 1$, or equivalently $|b_1| \approx 0$, so we can search outward from $b_1 = 0$ in the sign direction suggested by Appendix A.3.4. We write:

$$b_1 = sx, \qquad x \geq 0 \tag{38}$$

where $s$ denotes the chosen sign. Then the remaining variables for $i < n$ are recovered by:

$$b_i(x) = \frac{\sigma_1}{\sigma_i} sx, \qquad a_i(x) = \sqrt{1 - b_i(x)^2} \tag{39}$$

where the positive branch is used for $a_i$ as justified by Appendix A.3.5. The final phase is then recovered from the determinant constraint:

$$z_n(x) = \rho \prod_{i=1}^{n-1} z_i(x)^* \tag{40}$$

All $|b_i|$ are trivially bounded between 0 and 1 by the unit constraint. With the stationarity relation, it is implied that $x \leq \sigma_i/\sigma_1$ for every $i < n$, so the tightest generic bound is $x \leq \sigma_n/\sigma_1$. We can sharpen this further using $b_n$, which corresponds to the largest phase deviation in the optimal ordering. If $\Re(\rho) > 0$, then $|b_n|$ cannot be larger than $|\Im(\rho)|$ without violating the determinant constraint or being a suboptimal solution by Appendix A.3.4. If $\Re(\rho) \leq 0$, then the general bound remains as $|b_n| \leq 1$. Thus, it suffices to search for the optimal $x$ over the interval:

$$0 \leq x \leq \frac{\sigma_n}{\sigma_1}\beta(\rho), \qquad \beta(\rho) = \begin{cases} |\Im(\rho)|, & \Re(\rho) > 0 \\ 1, & \Re(\rho) \leq 0 \end{cases} \tag{41}$$

that maximizes the scalar function:

$$F(x) = 2\sum_{i=1}^{n} \sigma_i \Re(z_i(x)) \tag{42}$$

Several numerical strategies could be employed to solve the problem. In our implementation, we define the residual function:

$$r(x) = \frac{\sigma_1}{\sigma_n} sx - \Im(z_n(x)) \tag{43}$$

and treat the task as a root-finding problem to locate stationary point candidates within the interval. We find it sufficient to perform a coarse grid search on Eq. (43) over the interval to check for sign changes before locally searching within relevant brackets for the best optimizer. This strategy typically obtains the same solution as the analytical approach in lower dimensions while being much more computationally scalable, enabling the practical computation of solutions in higher dimensions.

### 3.3 Special Cases

We highlight some specific cases that require special treatment.

#### 3.3.1 Dimension $n \leq 2$

The case of $n = 1$ is trivial as there is only one element in $SU(1)$, the 1x1 matrix containing the scalar 1. The case of $n = 2$ is of significance as it is the highest dimension for which a general closed-form solution exists. It is also of special interest due its geometric interpretation as rotations of the sphere $S^2$ in 3D. We can write the closed-form expression as (see Appendix B.1 for derivation):

$$\begin{gathered} \eta = \sqrt{\sigma_1^2 + 2\sigma_1\sigma_2\Re(\rho) + \sigma_2^2} \\ z_1 = \frac{\sigma_1 + \sigma_2\rho}{\eta}, \qquad z_2 = \frac{\sigma_1\rho + \sigma_2}{\eta} \\ Q = U\,\mathrm{diag}(z_1^*, z_2^*)V^H \end{gathered} \tag{44}$$

The $n = 2$ case is also special because $SU(2)$ is isomorphic to the group of unit quaternions. This gives us a compact algebraic formula directly in terms of $M$ (see Appendix B.2 for derivation):

$$Q = \frac{M + \mathrm{adj}(M)^H}{\sqrt{||M||_F^2 + 2\Re(\det(M))}} \tag{45}$$

Both expressions become degenerate when $\rho = -1$ and $\sigma_1 = \sigma_2$, in which case the solution is not unique and we may choose the solution $z_1 = 1, z_2 = -1$.

#### 3.3.2 Singular Matrices

The general solution assumes that $M$ is nonsingular, i.e. all singular values are strictly positive. If $\mathrm{rank}(M) = n - 1$, then the construction in Section 2.2 is optimal. In this case, the zero singular direction carries no objective cost, so the determinant constraint can be satisfied entirely by setting $z_n = \rho$ while choosing $z_i = 1$ for $i < n$ to maximize the objective. If $\mathrm{rank}(M) \leq n - 2$, the optimizer is no longer unique. At least two zero singular directions are available, so any choice of phases satisfying $z_{n-1}z_n = \rho$ gives the same objective value while keeping $z_i = 1$ on the positive singular directions. In this case, the solution from Section 2.2 remains a valid solution.

| Trials | Setting | Naive I | Naive II | Linear Algebra | Algebraic | Numerical |
|---|---|---|---|---|---|---|
| 10000 | generic | 2.649 / 2.643 | 2.522 / 2.524 | **2.470 / 2.472** | **2.470 / 2.472** | **2.470 / 2.472** |
| 10000 | real det. | 2.789 / 2.787 | **2.499 / 2.485** | **2.499 / 2.485** | **2.499 / 2.485** | **2.499 / 2.485** |
| 10000 | repeated $\sigma$ | **3.462 / 3.459** | 3.682 / 3.702 | **3.462 / 3.459** | **3.462 / 3.459** | **3.462 / 3.459** |
| 10000 | rep. $\sigma$, real det. | **3.614 / 3.635** | **3.614 / 3.635** | **3.614 / 3.635** | **3.614 / 3.635** | **3.614 / 3.635** |
| 10000 | rank 1 | 2.642 / 2.628 | **2.223 / 2.214** | **2.223 / 2.214** | **2.223 / 2.214** | **2.223 / 2.214** |

(a) $SU(2)$ closed-form comparisons

| $n$ | Trials | Setting | Naive I | Naive II | Analytical NP | Analytical MP | Numerical |
|---|---|---|---|---|---|---|---|
| 3 | 1000 | generic | 3.791 / 3.800 | 3.691 / 3.690 | **3.647 / 3.665** | **3.647 / 3.665** | **3.647 / 3.665** |
| 4 | 100 | generic | 5.040 / 5.091 | 4.981 / 5.082 | 4.960 / 5.033$^\dagger$ | **4.948 / 5.019** | **4.948 / 5.019** |
| 5 | 10 | generic | 6.579 / 6.488 | 6.527 / 6.496 | **6.491 / 6.453** | **6.491 / 6.453** | **6.491 / 6.453** |
| 6 | 3 | generic | 8.096 / 7.799 | 8.008 / 7.729 | –$^\dagger$ | **7.989 / 7.719** | **7.989 / 7.719** |
| 7 | 1 | generic | 9.389 / 9.389 | 9.388 / 9.388 | –$^\dagger$ | **9.377 / 9.377** | **9.377 / 9.377** |
| 10 | 1000 | generic | 13.778 / 13.778 | 13.762 / 13.768 | – | – | **13.748 / 13.751** |
| 20 | 100 | generic | 28.909 / 28.870 | 28.905 / 28.864 | – | – | **28.899 / 28.863** |
| 1024 | 10 | generic | 1645.15 / 1645.15 | 1645.15 / 1645.15 | – | – | **1645.15 / 1645.15** |
| 4 | 100 | real det. | 5.193 / 5.271 | 5.023 / **5.018** | **5.020 / 5.018** | **5.020 / 5.018** | **5.020 / 5.018** |
| 4 | 100 | repeated $\sigma$ | 5.571 / 5.616 | 5.552 / 5.623 | 5.521 / **5.567** | **5.480 / 5.567** | **5.480 / 5.567** |
| 4 | 100 | rep. $\sigma$, real det. | 5.720 / 5.602 | 5.598 / 5.550 | **5.568 / 5.534** | **5.568 / 5.534** | **5.568 / 5.534** |
| 5 | 10 | rep. $\sigma$, real det. | 7.161 / **7.091** | 7.092 / **7.091** | **7.076 / 7.091** | **7.076 / 7.091** | **7.076 / 7.091** |
| 10 | 1000 | rep. $\sigma$, real det. | 14.050 / 14.033 | 14.012 / 14.014 | – | – | **14.003 / 14.004** |
| 4 | 100 | rank $n-1$ | 5.220 / 5.295 | **4.994 / 4.985** | **4.994 / 4.985** | **4.994 / 4.985** | **4.994 / 4.985** |
| 10 | 1000 | rank $n-1$ | 13.788 / 13.805 | **13.737 / 13.757** | – | – | **13.737 / 13.757** |
| 4 | 100 | rank $\leq n-2$ | 5.084 / 5.072 | **4.946 / 4.956** | **4.946 / 4.956** | **4.946 / 4.956** | **4.946 / 4.956** |
| 10 | 1000 | rank $\leq n-2$ | 13.867 / 13.860 | **13.815 / 13.800** | – | – | **13.815 / 13.800** |

(b) General solution comparisons

Table 1: Accuracy comparison across experimental settings. Each method entry reports mean/median Frobenius loss, with the lowest mean and median in each row bolded independently. ($^\dagger$) indicates that the method did not produce a valid output for one or more trials.

### 3.3.3 Real Determinant

If $\rho$ is real, then $\rho = \pm 1$. When $\rho = 1$, the optimal $Z$ is trivially the identity matrix since it yields the maximum possible value of Eq. (22) under the unit-norm constraints and satisfies the determinant constraint. The case $\rho = -1$ is more complicated. Multiple optimizers may exist as in the $SU(2)$ case discussed in Section 3.3.1. In addition, $b_1 = 0$ is always a valid stationary point in this case, which may cause issues for optimization schemes initialized at that point. Therefore, any search strategy for this special case should numerically handle $b_1 = 0$ appropriately as a candidate solution.

### 3.3.4 Repeated Singular Values

Although our derived solution does not assume the uniqueness of singular values, the presence of repeated singular values can cause issues in practice, especially when combined with a real determinant. We observe that this case potentially leads to multiple solutions or numerically unstable computations, and it is therefore a stress test for the methods evaluated in the next section.

## 4 Numerical Experiments

To validate our methods, we conduct simulated numerical experiments across varying matrix dimensions $n$. For each $n$, we generate a random $n \times n$ complex matrix whose real and imaginary parts are drawn uniformly from $[-2, 2]$, then project it onto $SU(n)$ using four approaches: the two naive baseline methods, our analytical solution, and our numerical solution. For $n = 2$, we replace the general solutions with our dedicated closed-form expressions, and for $n > 7$, we omit the analytical solution entirely due to computational intractability.

For the analytical solution, we test two root-finding backends: NumPy's built-in solver (denoted **NP**) and Aberth's method via the `mpmath` library at 100-digit precision (denoted **MP**), with the latter serving as a high-precision reference for theoretical validation. Reference code provided at https://github.com/akschion/Nearest_SU.

| Trials | Naive I | Naive II | Linear Algebra | Algebraic | Numerical |
|---|---|---|---|---|---|
| 10000 | 0.014 / 0.012 | 0.017 / 0.015 | 0.020 / 0.018 | 0.009 / 0.008 | 0.138 / 0.127 |

(a) $SU(2)$ timing comparisons

| $n$ | Trials | Naive I | Naive II | Analytical NP | Analytical MP | Numerical |
|---|---|---|---|---|---|---|
| 3 | 1000 | 0.010 / 0.010 | 0.015 / 0.013 | 0.362 / 0.347 | 737.113 / 787.313 | 0.194 / 0.183 |
| 4 | 100 | 0.010 / 0.010 | 0.014 / 0.014 | 0.838 / 0.840 | 1368.449 / 1323.908 | 0.194 / 0.184 |
| 5 | 10 | 0.054 / 0.016 | 0.017 / 0.015 | 12.900 / 12.583 | 24240.370 / 24109.011 | 0.626 / 0.360 |
| 10 | 1000 | 0.008 / 0.008 | 0.013 / 0.012 | – | – | 0.144 / 0.132 |
| 20 | 100 | 0.058 / 0.013 | 0.026 / 0.020 | – | – | 0.183 / 0.147 |
| 1024 | 10 | 100.569 / 101.591 | 148.820 / 136.331 | – | – | 164.045 / 155.652 |

(b) General timing comparisons

Table 2: Runtime comparison across generic experimental settings. Each method entry reports mean/median time in milliseconds. For $n = 2$, timings include SVD for SVD-based methods, while the algebraic method operates on the matrix directly. For $n > 2$, timings exclude the shared SVD operation.

**Results** Table 1 reports the performance of all methods across various experimental settings. Our proposed methods consistently outperform the naive baselines, which only match our methods in limited special cases. More specifically, for $n = 2$, our closed-form $SU(2)$ expressions and numerical method attain the best solution in all experiments. For $n \geq 3$, the analytical MP and numerical method achieve the best result whenever they are applied. The analytical NP method is reliable in the case $n = 3$, but begins to suffer from numerical instability in the root-finding step even at moderate dimensions such as $n = 4$. Both analytical methods, however, become computationally impractical in higher dimensions, whereas the numerical method retains strong performance in this regime.

Table 2 presents sample timings for each method. For $n = 2$, the algebraic expression is quite efficient as it is compact and does not require SVD. For $n \geq 3$, the MP method is a significant outlier, taking several orders of magnitude longer than the other approaches. In comparison, the NP method is much faster but remains limited by the cost of polynomial construction and root-finding. As expected, both naive methods are computationally inexpensive. Notably, our numerical method is a scalable and cost-effective approach for recovering the best solution, approaching comparable runtimes with the naive baselines in very high dimensions.

## 5 Conclusion

In this work, we have provided a complete characterization of the nearest special unitary matrix problem under the Frobenius norm. We demonstrated that while the $O(n)$, $U(n)$, and $SO(n)$ projections admit simple solutions involving SVD, the $SU(n)$ case is fundamentally more complex due to the continuous nature of the complex determinant constraint. By reducing the matrix nearness problem to a constrained phase optimization, we derived an exact analytical solution via polynomial elimination and a scalable numerical method based on univariate root-finding.

Our experiments confirm that common heuristics such as adjusting the global phase or correcting the smallest singular value consistently yield suboptimal results. In contrast, our proposed methods provide the exact projection in all dimensions, offering a reliable subroutine for applications in quantum gate synthesis, complex neural networks, and manifold optimization. These results fill a notable gap in the matrix nearness literature, providing both theoretical clarity and a practical computational framework for the $SU(n)$ projection.

## References

Akshay Chandrasekhar. Special unitary parameterized estimators of rotation. In *International Conference on Learning Representations (ICLR)*, 2026.

Nicholas J. Higham. Matrix nearness problems and applications. In M. J. C. Gover and S. Barnett, editors, *Applications of Matrix Theory*, pages 1–27. Oxford University Press, 1989.

Joseph B. Keller. Closest unitary, orthogonal and hermitian operators to a given operator. *Mathematics Magazine*, 48(4):192–197, 1975. ISSN 0025570X, 19300980.

K. B. Petersen and M. S. Pedersen. The matrix cookbook, November 2012.

Simone Romiti. $\mathrm{SU}(n)$ lattice gauge theories with physics-informed neural networks. *Phys. Rev. D*, 113:054511, Mar 2026.

Grace Wahba. A least squares estimate of satellite attitude. *SIAM Rev.*, 7(3):409, July 1965. ISSN 0036-1445.

Roeland Wiersema, Dylan Lewis, David Wierichs, Juan Carrasquilla, and Nathan Killoran. Here comes the su(n): multivariate quantum gates and gradients. *Quantum*, 8:1275, March 2024.

# On the Nearest Special Unitary Matrix

## Appendix

## A Derivation Details

### A.1 Expansion of Squared Frobenius Norm

Starting from the original objective:

$$\begin{aligned} &\min_{Q \in SU(n)} ||M - Q||_F^2 \\ &= \min_{Q \in SU(n)} Tr((M-Q)^H (M-Q)) \\ &= \min_{Q \in SU(n)} Tr(M^H M - Q^H M - M^H Q + Q^H Q) \\ &= \min_{Q \in SU(n)} Tr(M^H M) - Tr(Q^H M) - Tr(M^H Q) + Tr(Q^H Q) \\ &= \min_{Q \in SU(n)} Tr(M^H M) - 2\Re(Tr(Q^H M)) + Tr(Q^H Q). \end{aligned}$$

by linearity of trace.

### A.2 Optimality Condition from Lagrange Multipliers

Using the method of Lagrange multipliers for optimization, we define the following Lagrangian:

$$\mathcal{L} = 2\Re(Tr(Q^H M)) - \Re(Tr(\Lambda(Q^H Q - I))) - 2\Re(\lambda(\det(Q) - 1))$$

where $\Lambda$ is a Hermitian matrix of Lagrange multipliers and $\lambda$ is a complex scalar multiplier. The scalar coefficients in front of the constraint terms are immaterial to the optimizer and are chosen only for convenience. Using standard complex matrix differentiation rules (Petersen and Pedersen, 2012):

$$\frac{\partial \mathcal{L}}{\partial Q} = M^* - \frac{1}{2} Q^* \Lambda^T - \frac{1}{2} Q^* \Lambda^* - \lambda \det(Q)(Q^{-1})^T = 0$$

$$\frac{\partial \mathcal{L}}{\partial Q^*} = M - \frac{1}{2} Q\Lambda - \frac{1}{2} Q\Lambda^H - \lambda^* \det(Q^*)((Q^*)^{-1})^T = 0$$

Together with the constraints

$$Q^H Q = I, \qquad \det(Q) = 1$$

these define the stationary system. Since $\Lambda$ is Hermitian, $\Lambda^T = \Lambda^*$. Also, for $Q \in SU(n)$ we have $\det(Q) = \det(Q^*) = 1$, $Q^{-1} = Q^H$, $(Q^{-1})^T = Q^*$, and $((Q^*)^{-1})^T = Q$. The first-order conditions therefore simplify to:

$$\frac{\partial \mathcal{L}}{\partial Q} = M^* - Q^* \Lambda^* - \lambda Q^* = 0$$

$$\frac{\partial \mathcal{L}}{\partial Q^*} = M - Q\Lambda - \lambda^* Q = 0$$

These two equations are conjugates of one another, so it suffices to work with the second. Rearranging gives:

$$M = Q(\Lambda + \lambda^* I)$$

$$Q^H M = \Lambda + \lambda^* I$$

This is the stationary relation stated in Eq. (14).

### A.3 Proof Arguments

General arguments for various properties of the solutions presented.

### A.3.1 Nearest Special Unitary Matrix to Projective Matrix

For projective matrix $M$, the equivalence relation is defined such that:

$$M \sim \gamma M, \qquad \gamma \in \mathbb{C}, \gamma \neq 0$$

Let $\gamma = |\gamma| e^{i\phi}$. The nearest unitary matrix to $\gamma M$ is $e^{i\phi} A$, where $A$ is the polar factor of $M$. Thus, the magnitude of $\gamma$ has no effect on the optimal unitary projection. Scaling $M$ by a phase then is a unitary transformation. By the unitary invariance of the Frobenius norm, the distance between $\gamma M$ and the nearest unitary matrix is constant under phase scaling. Crucially, $SU(n)$ is a subgroup of $U(n)$, and the intersection of the set of nearest unitary matrices $\{e^{i\phi} A\}$ with the $SU(n)$ manifold occurs only when $e^{i\phi} = \det(A)^{-\frac{1}{n}}$ so that $\det(e^{i\phi} A) = 1$, the only valid determinant for special unitary matrices. By selecting this specific phase, the unitary lower bound for the entire projective class is achieved, making it the unique point where the optimal unitary and special unitary projections coincide.

### A.3.2 Degree of the Univariate Polynomial

The product in Eq. (31) contains $n-1$ factors, so the initial equation has degree $n-1$. Moreover, it is affine in each real part $a_i$ individually. Thus, before eliminating $a_i$, the current equation can be written as:

$$c_{i,1} a_i + c_{i,0} = 0$$

where $c_{i,1}$ and $c_{i,0}$ do not contain $a_i$.

The unit-norm relations in Eq. (32) allow all higher powers of an eliminated variable to be reduced to either a constant term or a linear term in that variable. Eliminating $a_i$ is then done by squaring the affine relation and substituting:

$$a_i^2 = 1 - \frac{\sigma_n^2}{\sigma_i^2} b_n^2$$

This doubles the degree of the current equation in the remaining variables. Equivalently, if $d_k$ denotes the degree after $k$ eliminations, the iterative construction satisfies:

$$d_{k+1} = 2d_k, \qquad d_0 = n-1$$

Repeating the step for $a_1, \ldots, a_{n-1}$ gives:

$$\deg(f) = 2^{n-1}(n-1)$$

### A.3.3 Degree Reduction for Even Dimensions

Write the initial determinant equation after substituting Eq. (29) as:

$$P(a_1, \ldots, a_{n-1}, b_n) = \Im\left(\rho \prod_{i=1}^{n-1} \left(a_i - \frac{\sigma_n}{\sigma_i} b_n i\right)\right) - b_n$$

If $n$ is even, then $n-1$ is odd. Under the simultaneous sign change:

$$(a_1, \ldots, a_{n-1}, b_n) \mapsto (-a_1, \ldots, -a_{n-1}, -b_n)$$

each factor $a_i - \frac{\sigma_n}{\sigma_i} b_n i$ is multiplied by $-1$, and hence the imaginary component of the product is multiplied by $(-1)^{n-1}$. Therefore, for even $n$, we can deduce that $P$ is odd:

$$P(-a_1, \ldots, -a_{n-1}, -b_n) = -P(a_1, \ldots, a_{n-1}, b_n)$$

Note that if a function is odd or even, then every term has odd or even total degree respectively.

Now consider a single elimination step. After reducing powers of $a_i$ with the parity-preserving substitution:

$$a_i^2 = 1 - \frac{\sigma_n^2}{\sigma_i^2} b_n^2$$

the current equation has the affine form from Eq. (33):

$$c_{i,1}a_i + c_{i,0} = 0$$

where $c_{i,1}$ and $c_{i,0}$ do not contain $a_i$. If the current equation has definite parity under a simultaneous sign change of its variables, then the two terms $c_{i,1}a_i$ and $c_{i,0}$ have the same parity. Since $a_i$ is odd under this sign change, $c_{i,1}$ and $c_{i,0}$ have opposite parity. Eliminating $a_i$ as in Eq. (34) yields:

$$c_{i,1}^2 \left(1 - \frac{\sigma_n^2}{\sigma_i^2} b_n^2\right) - c_{i,0}^2 = 0$$

Since $c_{i,1}^2$, $c_{i,0}^2$, and the factor $1 - \frac{\sigma_n^2}{\sigma_i^2} b_n^2$ are all even, the resulting equation is even as well.

Initially, $P$ and its affine form in $a_1$ are odd for even $n$. The first elimination step therefore produces an even equation. Repeating the elimination process for $a_2, \ldots, a_{n-1}$ keeps the equation even after every elimination. Thus, the final univariate polynomial in $b_n$ is even as well, satisfying $f(-b_n) = f(b_n)$. Equivalently, this means the roots of $f$ occur in symmetric pairs $\pm b_n$, and there exists a polynomial $g$ such that $f(b_n) = g(b_n^2)$. As a result, the polynomial root-finding step may be performed in the variable $\Delta = b_n^2$. Viewing the polynomial as a function of $\beta$ halves the degree count from Appendix A.3.2, giving us:

$$\deg(g(\Delta)) = \frac{1}{2} \deg(f(b_n)) = 2^{n-2}(n-1)$$

### A.3.4 Sign of $b_i$

For nonsingular matrices with $\Im(\rho) \neq 0$, the stationarity relation in Eq. (29) implies that all imaginary parts $b_i$ have the same nonzero sign. Write $z_i = e^{i\theta_i}$ with principal angles $\theta_i \in (-\pi, \pi]$. Then all $\theta_i$ have the same sign as the $b_i$, and the determinant constraint can be phrased as:

$$\sum_{i=1}^{n} \theta_i = \arg(\rho) \pmod{2\pi}$$

The objective:

$$2 \sum_{i=1}^{n} \sigma_i \cos(\theta_i)$$

is maximized by keeping the angles $\theta_i$ as close to zero as possible since cosine is strictly decreasing as $|\theta_i|$ increases on $(-\pi, \pi]$.

Now suppose $\Im(\rho) > 0$ and let $\phi = \arg(\rho) \in (0, \pi)$. Any feasible branch has:

$$S \coloneqq \sum_{i=1}^{n} \theta_i = \phi + 2\pi k \tag{46}$$

for some integer $k$. The branch $S = \phi$ has the smallest magnitude. If a candidate uses any other branch, then $|S| > \phi$. Since all $\theta_i$ have the same sign, define:

$$\tilde{\theta}_i = \frac{\phi}{S}\theta_i$$

Then $\sum_i \tilde{\theta}_i = \phi$, so the phases $\tilde{z}_i = e^{i\tilde{\theta}_i}$ satisfy the same determinant constraint. Moreover:

$$|\tilde{\theta}_i| = \frac{\phi}{|S|}|\theta_i| \leq |\theta_i| \tag{47}$$

for every $i$, so $\cos(\tilde{\theta}_i) \geq \cos(\theta_i)$. Thus, the maximum can be seen to occur at equality where $S = \phi$. On this optimal branch, the angles are positive, and therefore $b_n > 0$. The same reasoning applies with signs reversed when $\Im(\rho) < 0$, implying:

$$\operatorname{sign}(b_n) = \operatorname{sign}(\Im(\rho))$$

In the case of singular matrices or $\Im(\rho) = 0$, a degeneracy exists where the solution is typically trivial or non-unique.

### A.3.5 Sign of $a_i$

If $a_i < 0$, then $|\theta_i| > \frac{\pi}{2}$. Consequently, if two indices $i \neq j$ satisfy $a_i < 0$ and $a_j < 0$, then

$$|\theta_i| + |\theta_j| > \pi$$

so the determinant phase is being reached by moving more than halfway around the unit circle. By Appendix A.3.4, this cannot be optimal since the same target phase can be attained by taking the shorter path ($|\theta_i| \leq |\arg \rho|$) where the angles are closer to zero. Hence, at most one $a_i$ can be negative.

Now suppose exactly one $a_i$ is negative. With singular values ordered $\sigma_1 \geq \cdots \geq \sigma_n$, the negative $a_i$ which is smallest should be optimally assigned to the smallest singular value by the rearrangement inequality. Therefore, if a negative $a_i$ occurs at all, it can only occur at index $n$, i.e. $a_i \geq 0$ for $i < n$.

## B $SU(2)$ Formula Derivations

### B.1 Linear Algebra Expression

Since there are only two dimensions, the determinant constraint immediately removes one phase variable:

$$z_2 = \rho z_1^*$$

where $\rho = \det(U)\det(V^H)$. Let $z_1 = a_1 + b_1 i$. Then:

$$z_2 = (a_1\Re(\rho) + b_1\Im(\rho)) + (a_1\Im(\rho) - b_1\Re(\rho))i$$

so the phase objective reduces to maximizing a linear function over the unit circle:

$$\max_{a_1,b_1} \ (\sigma_1 + \sigma_2\Re(\rho))a_1 + \sigma_2\Im(\rho)b_1 \qquad s.t. \qquad a_1^2 + b_1^2 = 1$$

Define:

$$\eta = \sqrt{(\sigma_1 + \sigma_2\Re(\rho))^2 + \sigma_2^2\Im(\rho)^2} = \sqrt{\sigma_1^2 + 2\sigma_1\sigma_2\Re(\rho) + \sigma_2^2}$$

Assuming $\eta \neq 0$, the maximizer is simply the normalized coefficient vector:

$$a_1 = \frac{\sigma_1 + \sigma_2\Re(\rho)}{\eta}, \qquad b_1 = \frac{\sigma_2\Im(\rho)}{\eta}$$

Therefore:

$$z_1 = \frac{\sigma_1 + \sigma_2\rho}{\eta}$$

$$z_2 = \frac{\rho(\sigma_1 + \sigma_2\rho^*)}{\eta} = \frac{\sigma_1\rho + \sigma_2}{\eta}$$

Thus, the solution is a simple expression involving the SVD elements. The only degenerate case occurs when $\eta = 0$ which requires $\sigma_1 = \sigma_2$ and $\rho = -1$. In that setting, the linear objective vanishes for every feasible $z_1$, so the optimizer is not unique.

### B.2 Algebraic Expression

Utilizing the isomorphism between $SU(2)$ and quaternions, we can directly express the solution in terms of $M$ without performing SVD. We can choose an isomorphism between a quaternion $q$ and $Q \in SU(2)$ such that:

$$q = w + xi + yj + zk, \qquad w^2 + x^2 + y^2 + z^2 = 1, \ \ w, x, y, z \in \mathbb{R}$$

$$Q = \begin{bmatrix} w + xi & y + zi \\ -y + zi & w - xi \end{bmatrix}$$

Now write:

$$M = \begin{bmatrix} m_{11} & m_{12} \\ m_{21} & m_{22} \end{bmatrix}$$

We can maximize the trace objective in Eq. (13) directly. Expanding in terms of the entries of $M$ gives:

$$\begin{aligned}\Re(Tr(Q^H M)) &= \Re\big((w - xi)m_{11} + (y - zi)m_{12} + (-y - zi)m_{21} + (w + xi)m_{22}\big) \\ &= w\Re(m_{11} + m_{22}) + x\Im(m_{11} - m_{22}) + y\Re(m_{12} - m_{21}) + z\Im(m_{12} + m_{21})\end{aligned}$$

Thus the problem is the maximization of a linear function over $S^3$:

$$\max_{w,x,y,z} \quad c_w w + c_x x + c_y y + c_z z \qquad s.t. \qquad w^2 + x^2 + y^2 + z^2 = 1$$

where:

$$\begin{aligned} c_w &= \Re(m_{11} + m_{22}), \qquad & c_x &= \Im(m_{11} - m_{22}) \\ c_y &= \Re(m_{12} - m_{21}), \qquad & c_z &= \Im(m_{12} + m_{21}) \end{aligned}$$

Define the normalizer:

$$\gamma = \sqrt{c_w^2 + c_x^2 + c_y^2 + c_z^2}$$

Assuming $\gamma \neq 0$, the maximizer is the normalized coefficient vector:

$$w = \frac{c_w}{\gamma}, \qquad x = \frac{c_x}{\gamma}, \qquad y = \frac{c_y}{\gamma}, \qquad z = \frac{c_z}{\gamma}$$

Substituting these coefficients back into the matrix form of $Q$ gives:

$$Q = \frac{1}{\gamma}\begin{bmatrix} m_{11} + m_{22}^* & m_{12} - m_{21}^* \\ m_{21} - m_{12}^* & m_{22} + m_{11}^* \end{bmatrix} = \frac{M + \mathrm{adj}(M)^H}{\gamma}$$

where $\mathrm{adj}(\cdot)$ is the adjugate. Finally, the normalization can be written directly in terms of matrix invariants:

$$\gamma^2 = |m_{11} + m_{22}^*|^2 + |m_{12} - m_{21}^*|^2 = ||M||_F^2 + 2\Re(\det(M))$$

Therefore, when $\gamma \neq 0$, the algebraic formula is:

$$Q = \frac{M + \mathrm{adj}(M)^H}{\sqrt{||M||_F^2 + 2\Re(\det(M))}}$$

If $\gamma = 0$, the linear objective vanishes over $S^3$ and the optimizer is not unique, matching the degenerate case in the linear algebra expression.